Introduction to Informatics

Module Description

Module code	CS101-F26	**Credits**	7 ECTS
Programme	Foundation programme	**Level**	Semester 1
Semester	Fall 2026	**Prerequisites**	None
Module leader	Dr. Ali Khudiyev	**Email**	ali.khudiyev@wosm.academy
Office hours	Monday–Friday, 10:00–17:00 (Baku time, UTC+4)		

1. Module Overview

Introduction to Informatics (ItI) is the first-semester lecture course introducing computer science and object-oriented programming. Each week the lecture introduces the concepts that are then supported with tests, quizzes, and other supplementary materials. The language of instruction and implementation is Java.

2. Aims

- TODO(owner): aims of the module.

3. Learning Outcomes

After completing this course, you will:

LO1. Understand the essential concepts of computer science and be able to apply them in a practical, scientifically grounded, and responsible way.

LO2. Learn key concepts such as algorithms, syntax and semantics, and how to evaluate program efficiency in terms of memory use and runtime.

LO3. Solve well-defined algorithmic problems and develop basic concurrent and distributed applications using Java or a similar object-oriented programming language. This foundation will help you learn other imperative and object-oriented programming languages independently.

4. Syllabus

Unit	Topic
1	Introduction
2	Control Structure
3	Data Types
4	Object Orientation I
5	Object Orientation II
6	Object Orientation III
7	Algorithms
8	Programming Languages
9	Graphical User Interfaces
10	Recursion
11	Syntax, Semantics, Verification
12	Beyond Programming
13	Course Review & Exam

5. Teaching and Learning Methods

The module is self-paced. All materials are provided on Moodle.

Activity	Hours
Lectures	TODO(owner)
Labs / Tutorials	TODO(owner)
Independent study	TODO(owner)
Total	**TODO(owner)**

6. Assessment

Your knowledge will be assessed throughout the course with quizzes, tutor and homework exercises, team project, and the final exam.

TODO(owner): weight of each assessment component and the learning outcomes it covers.

7. Reading List

- Sedgewick, R. and Wayne, K. *Computer Science: An Interdisciplinary Approach* (2016) and *Introduction to Programming in Java* (2017). Companion material: introcs.cs.princeton.edu.

8. Materials and Licence

Course materials are published under the Creative Commons Attribution-ShareAlike 4.0 licence unless stated otherwise. See wosm.academy/licence for details.

9. Policies

- **Attendance:** TODO(owner)
- **Late submissions:** TODO(owner)
- **Academic integrity:** All submitted work must be your own. See the Code of Conduct.